Lappeenrannan teknillinen yliopisto
School of Business and Management

Sofware Development Skills

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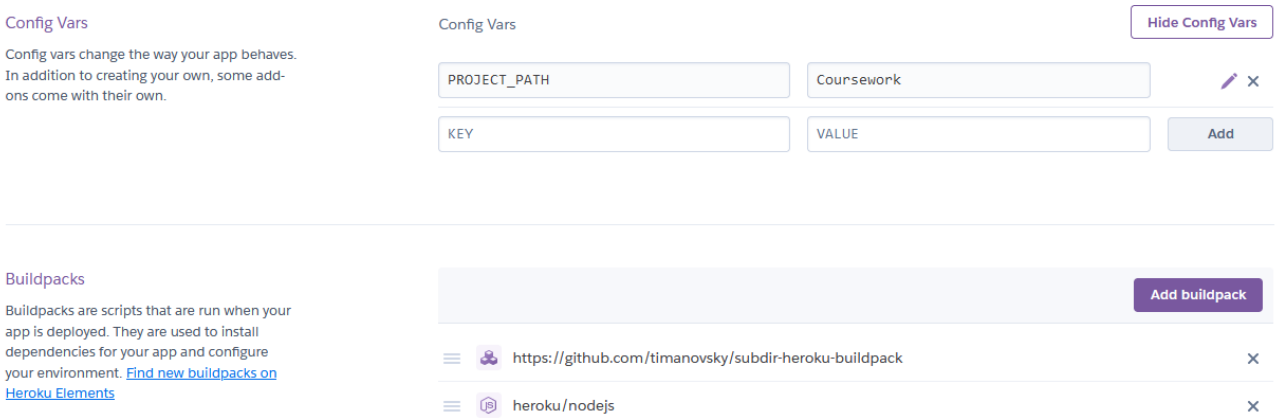
LEARNING DIARY, FULL-STACK MODULE

LEARNING DIARY

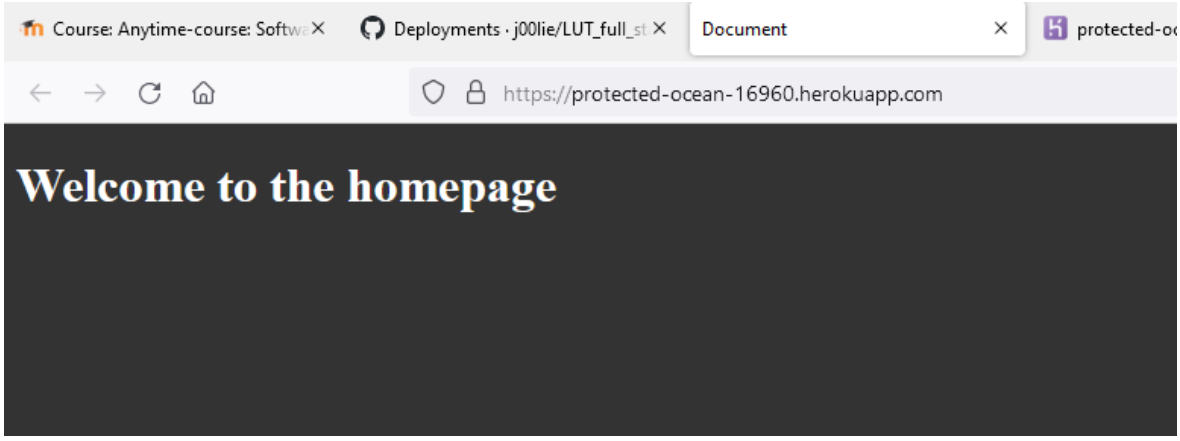
13.6.2022

Today I read through the administrative part of the course and completed the first section covering node.js basics. I already had Node setup on my machine, so I skipped that part. In addition, I've used Git previously and recently I've been using GitHub desktop due to its nice GUI and convenience. I coded along the demos regarding the basic tools of node.js such as the path, fs, url and http. I successfully deployed the server on my localhost using fs, path and http tools. In the latter part of this instructional the goal was to deploy the 'site' using Heroku. As I already had done the push to my remote, I opted to use Herokus integration with GitHub. Because my root folder in GitHub contains the 'Coursework' folder which in turn contains the actual server files, the deployment initially failed. I had to work around this by creating a configuration variable to point to my folder with the server files in addition to adding a custom buildpack.

Screenshot of custom buildpack and configuration variable:



Deployed site:



14.6.2022

Today I learned the basics of MongoDB. Prior to this I had only used relational SQL-databases, but luckily the operations and database elements in MongoDB were quite intuitive. In addition, I learned about setting up the MongoDB cloud environment. Screenshot of some of the database commands:

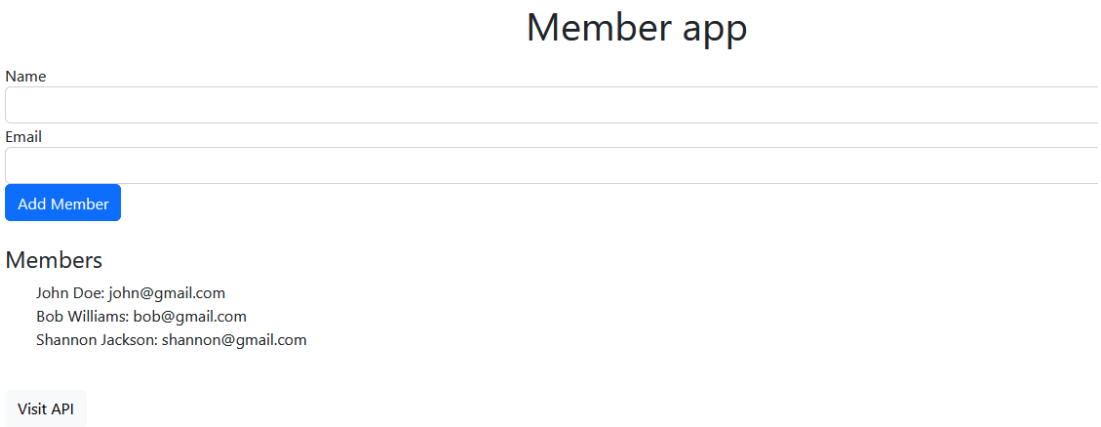
Coursework > mongoInsert.txt

```
db.posts.insert({
    title: 'Post one',
    body: 'Body of post one',
    categoy: 'News',
    likes: 4,
    tags: ['news', 'events'],
    user:{name:'John', status:'author'},
    date: Date()
})
db.posts.insertMany([
    {
    title: 'Post two',
    body: 'Body of post two',
    categoy: 'Technology',
    date: Date()
},{
    title: 'Post three',
    body: 'Body of post three',
    categoy: 'News',
    date: Date()
},{
    title: 'Post four',
    body: 'Body of post four',
    categoy: 'Entertainment',
    date: Date()
}])


db.posts.update({title: 'Post two'},
{
    title: 'Post  two',
    body: 'New post 2 body',
    date: Date()

},
{
    upsert: true
}
)
db.posts.update({title: 'Post two'},
{
    $set:{
        categoy: 'Technology'
    }

}

)
db.posts.update({title: 'Post one'},
{
    $set:{
        comments: [
            {
```

15.6.2022

This part of the course was a overview of the express.js framework. The basic usage was pretty straight forward and significantly more efficient compared to the server we built with pure Node.js in the first part. I also learned to use the postman app to handle http requests. In addition, the tutorial briefly covered the use of server side templates using express-handlebars.

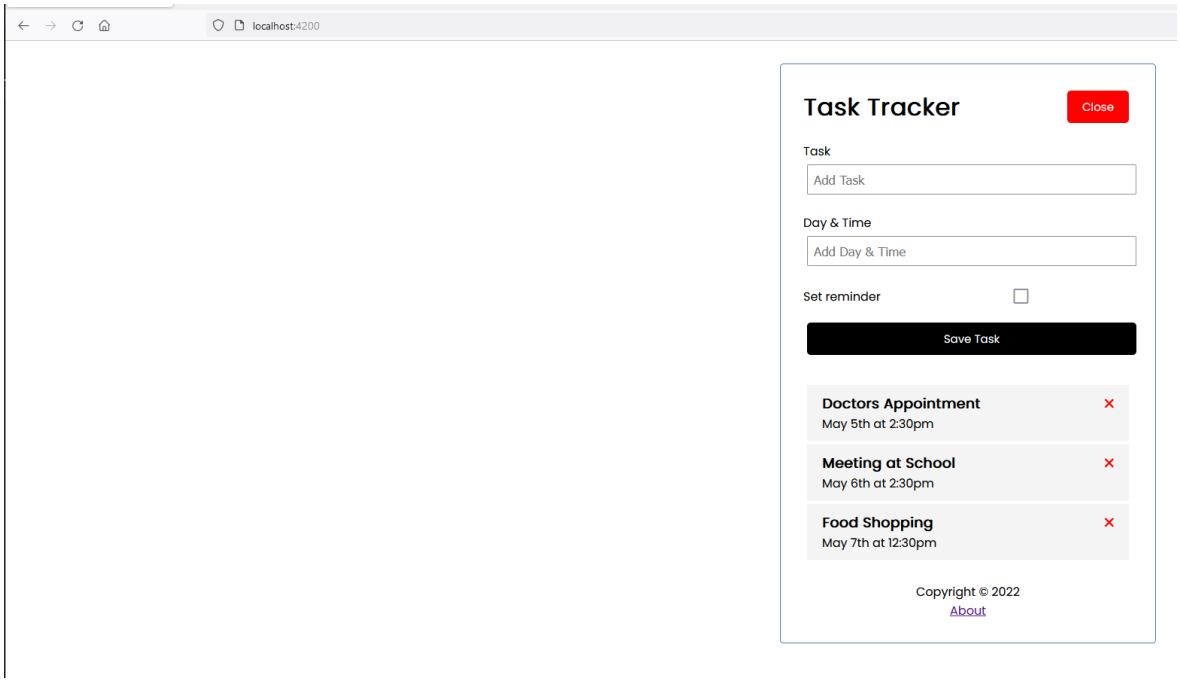
Screenshot:

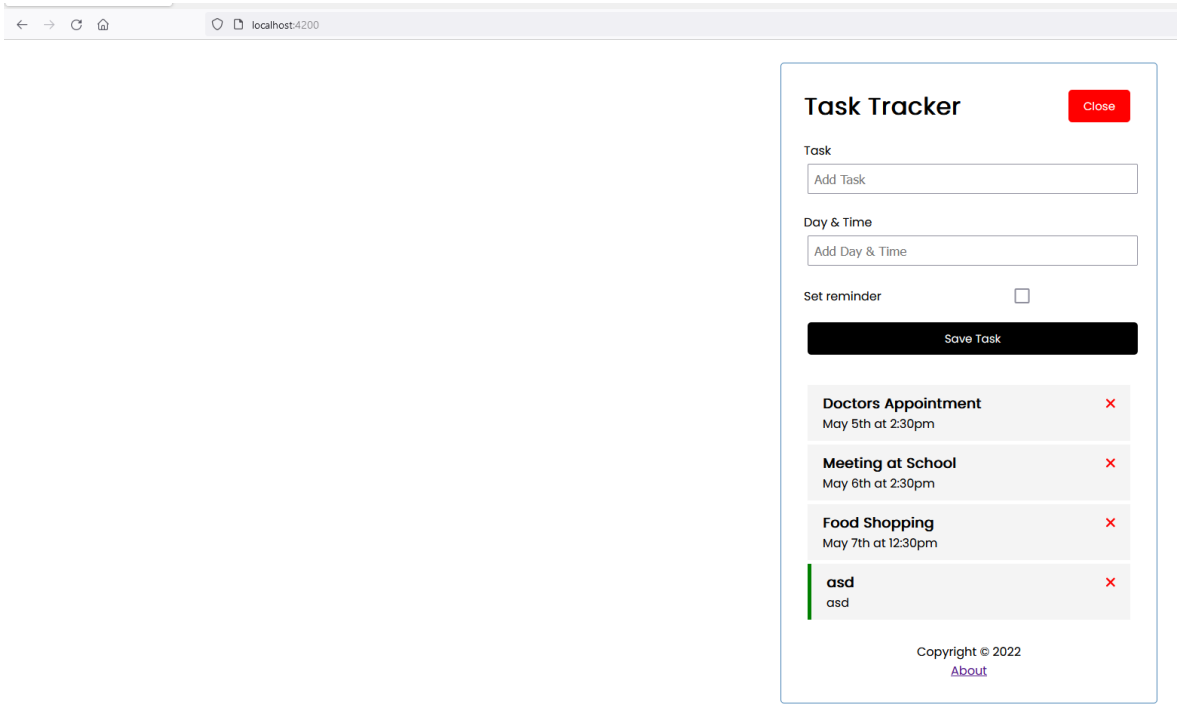


16.6-17.6.2022

I spent these days working on learning Angular. As a resource I used the Traversy [crash course] to Angular because the previous courses have excellent. This video goes through much of the same stuff provided in the Moodle exercise, but in more detail. I found this framework to be quite complicated in comparison with the previous ones. In addition, some of the TypeScript concepts were a bit confusing since I've had exposure to only normal JavaScript.

Screenshots:





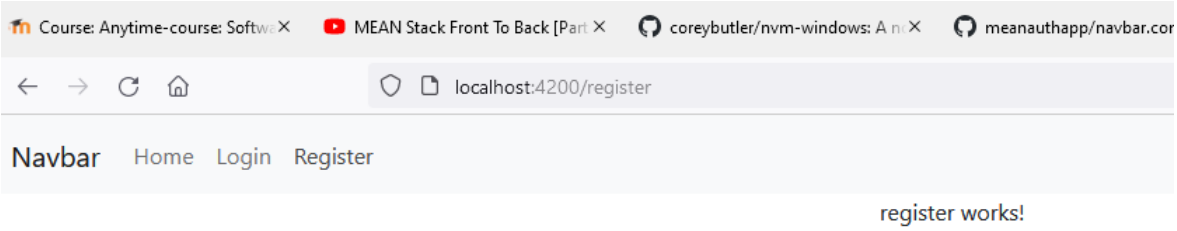
19-20.6.22

Got started with the MEAN-stack tutorial and built the backend using express. I had some initial problems with the jwt-payload but after some console logging I figured out that the user id (and the other stuff) is stored within a data-object so the id had to be accessed through jwt_payload.data._id.

I also began working on the frontend with angular. Iintially I tried to install the old version of Angular used in the videos but was unsuccessful due to my new version of node. I then

proceeded to uninstall all my node.js related things and installed the node version manager for windows, as it might come in handy in the future as well. With nvm I installed node 12 and was able to get the old version of angular working. Because the tutorial is 5 years old, I also had some problems with the CSS getting through. So I decided to do a slightly different navbar that was easily available from bootstrap. I also built the registering component with the validation. I was unable to get the flash message for the email validation to work, since all browsers have their built in function for this.

Screenshot:



21.6.22

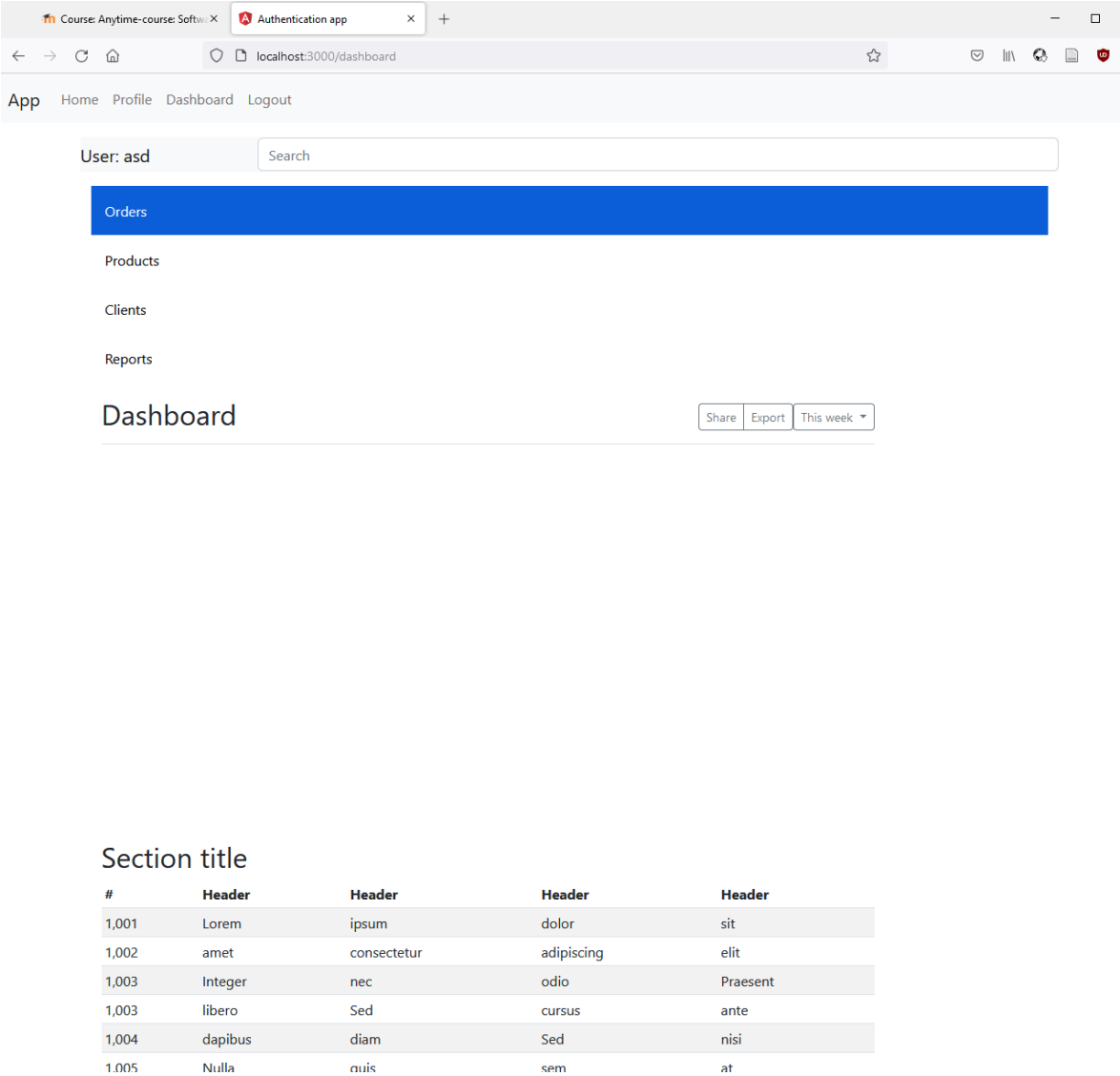
In todays tutorial we implemented the login with authentication and logout functionalities. I initially couldn't get the login to work and kept getting the error message indicating user wasn't found even though they were definitely in the database. The cause for this turned out to be a minor typo in the login typescript.

22.6.22

This part completed the route guarding and authorization. Everything went more or less smoothly along with the video. In the video the routes to profile, dashboard etc. were hidden with the same isLoggedIn()-function but I couldn't get this to work within the HTML so I made a separate isLoggedOut()-function to the typescript file that just returns the inverse of the original function. After getting the code ready we built the application with Angulars build method to the public folder.

28.6.22

Today I built a dashboard using a bootstrap template and added a responsive sidebar to it. The dashboard is designed to function as a interface for accounting purposes or possibly managing a online store. In the HTML there's a spot for graphing data but unfortunately, I did not have enough time to complete this functionality for this project. The graphing could

perhaps be done by creating a service and using it via the dashboard components typescript. In addition, I built the project using Angulars build method and now the project can be run through the initial backend port 3000.

Screenshot: